CONSTRUCTIBILITY IS TRANSITIVE FOR A SPACE WITH A BASIS OF QUASI-COMPACT OPENS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a constructible subset of such a space is retrocompact, and the description of a retrocompact subset as having a basis of quasi-compact opens. Invoked at the start of the counted proof (source lines 2988-2991). Gives the induced topology a basis of quasi-compact opens.
- The statement that the trace of a constructible subset on a constructible subset is constructible in the latter. Invoked inside the counted proof (source lines 2994-2999). Reduces the assertion to the single-piece case.
- The two pullback statements for a closed constructible image and for a quasi-compact open immersion. Invoked at the end of the counted proof (source lines 3003-3007). Handles the two inclusions of the factorization.

Source reference. Original lemma lemma-closed-constructible-image: sources/stacks-topology/pinned-original-source.tex, source lines 2916–2916. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-constructible-image, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-is-retrocompact: sources/stacks-topology/pinned-original-source.tex, source lines 2944–2944. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-is-retrocompact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-intersect-constructible-with-constructible: sources/stacks-topology/pinned-original-source.tex, source lines 2967–2967. Navigation pointer only: the counted text refers to the Stacks item lemma-intersect-constructible-with-constructible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-quasi-compact-open-immersion-constructible-image: sources/stacks-topology/pinned-original-source.tex, source lines 2837–2837. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-open-immersion-constructible-image, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. Assume X has a basis consisting of quasi-compact opens. Let E be constructible in X and $F \subset E$ constructible in E. Then F is constructible in X.*

Proof. Observe that any retrocompact subset T of X has a basis for the induced topology consisting of quasi-compact opens. In particular this holds for any constructible subset (Lemma lemma-constructible-is-retrocompact). Write $E = E_1 \cup \ldots \cup E_n$ with $E_i = U_i \cap V_i^c$ where $U_i, V_i \subset X$ are retrocompact open. Note that $E_i = E \cap E_i$ is constructible in E by Lemma lemma-intersect-constructible-with-constructible. Hence $F \cap E_i$ is constructible in E_i by Lemma lemma-intersect-constructible-with-constructible. Thus it suffices to prove the lemma in case $E = U \cap V^c$ where $U, V \subset X$ are retrocompact open. In this case the inclusion $E \subset X$ is a composition

$$E = U \cap V^c \to U \to X$$

Then we can apply Lemma lemma-closed-constructible-image to the first inclusion and Lemma lemma-quasi-compact-open-immersion-constructible-image to the second. □